Medical Environmental Health & Parasitology

Comprehensive Study Guide & Practice Questions (MCQ & True/False)

Part 1: Multiple Choice Questions (MCQ)

Ventilation, Waste and Water

1. The modern concept of ventilation includes:

A) Only removing stale air
B) Replacing vitiated air with fresh outdoor air and controlling quality of incoming air
C) Only cooling
D) Only adding humidity

Answer: **B**

2. Which is NOT controlled in the quality of incoming air?

A) Temperature
B) Humidity
C) CO2/Oxygen
D) Noise level

Answer: **D**

3. Which is NOT part of waste management?

A) Collection
B) Source separation
C) Advertising
D) Disposal

Answer: **C**

4. Which is NOT a health problem due to solid waste?

A) Skin disease
B) Gastro-intestinal disease
C) Liver disease
D) Cataract

Answer: **D**

5. Burning solid waste may cause:

A) Respiratory disease
B) Scurvy
C) Rickets
D) Goiter

Answer: **A**

6. Burning waste affects climate change and creates:

A) Pure water
B) Toxic ash
C) Fertilizer
D) Oxygen

Answer: **B**

7. The composition of water (H2O):

A) One H, two O
B) Two H, one O
C) Two H, two O
D) One H, one O

Answer: **B**

8. Hydrogen atoms are bonded to oxygen in water by:

A) Ionic bonds
B) Covalent bonds
C) Metallic bonds
D) Van der Waals forces

Answer: **B**

9. Potable water is water that is:

A) Colored
B) Safe for ingestion
C) Used only for washing
D) Salty

Answer: **B**

10. Maximum turbidity of drinking water:

A) 1 NTU
B) 5 NTU
C) 10 NTU
D) 50 NTU

Answer: **B**

11. Ideal turbidity of drinking water:

A) Below 1 NTU
B) 5 NTU
C) 10 NTU
D) Above 20 NTU

Answer: **A**

12. Acceptable pH range (per slides):

A) 1–5
B) 5–9
C) 9–14
D) 0–3

Answer: **B**

13. A pH above 9 constrains:

A) Filtration
B) Boiling
C) Chlorination
D) Precipitation

Answer: **C**

14. Zero coliform means the water is:

A) Fit
B) Probably contaminated
C) Contaminated
D) Toxic

Answer: **A**

15. 1–10 E. coli indicates:

A) Fit
B) Probable contamination
C) Contaminated
D) Sterile

Answer: **B**

16. 10–100 E. coli indicates:

A) Fit
B) Probable contamination
C) Contaminated
D) Excellent quality

Answer: **C**

17. Minimum water for drinking and domestic hygiene (Sphere 2018):

A) 5 L/person/day
B) 10 L
C) 15 L
D) 50 L

Answer: **C**

18. Maximum people per tap:

A) 50
B) 100
C) 250
D) 500

Answer: **C**

19. The tap standard is based on a flow rate of:

A) 7.5 L/min
B) 17 L/min
C) 15 L/min
D) 1 L/min

Answer: **A**

20. Maximum people per hand pump:

A) 100
B) 250
C) 500
D) 50

Answer: **C**

21. The hand pump standard is based on a flow rate of:

A) 7.5 L/min
B) 17 L/min
C) 25 L/min
D) 2 L/min

Answer: **B**

22. Maximum people per laundry facility:

A) 50
B) 100
C) 250
D) 500

Answer: **B**

23. Maximum people per bathing facility:

A) 50
B) 100
C) 250
D) 500

Answer: **A**

24. Chemical added in the first step of water treatment:

A) Aluminum sulphate
B) Sodium chloride
C) Calcium carbonate
D) Ammonia

Answer: **A**

25. Which is NOT a sterilization method mentioned?

A) Boiling
B) Chlorine
C) Ozone gas
D) Freezing

Answer: **D**

26. Which is NOT related to drinking-water contamination?

A) Cholera
B) Typhoid
C) Polio
D) Malaria

Answer: **D**

Food Contamination

27. Foodborne illnesses are caused by:

A) Bacteria only
B) Bacteria, viruses, parasites or chemicals
C) Viruses only
D) Noise

Answer: **B**

28. Which is NOT a bacterial foodborne pathogen?

A) Salmonella
B) Campylobacter
C) Enterohaemorrhagic E. coli
D) Norovirus

Answer: **D**

29. Salmonellosis outbreaks are usually linked to:

A) Eggs and poultry
B) Distilled water
C) Salt
D) Sugar

Answer: **A**

30. Campylobacter cases are mainly caused by:

A) Raw milk, undercooked poultry, drinking water
B) Canned fruit
C) Bread
D) Honey

Answer: **A**

31. Which infection can lead to miscarriage or death of newborns?

A) Listeria
B) Norovirus
C) Giardia
D) Cholera

Answer: **A**

32. Listeria can:

A) Not survive cold
B) Grow at refrigeration temperatures
C) Live only in water
D) Infect only plants

Answer: **B**

33. Foods implicated in cholera outbreaks:

A) Rice, vegetables, millet gruel, seafood
B) Only chocolate
C) Only bread
D) Only dairy

Answer: **A**

34. Common viral cause of foodborne infection with explosive vomiting:

A) Hepatitis A
B) Norovirus
C) Salmonella
D) Taenia

Answer: **B**

35. Hepatitis A virus can cause:

A) Long-lasting liver disease
B) Hearing loss
C) Cataract
D) Hypertension

Answer: **A**

36. Beef tapeworm is:

A) Taenia saginata
B) Giardia lamblia
C) Ascaris
D) Vibrio cholerae

Answer: **A**

37. Which parasite may infect through food or contact with animals?

A) Hymenolepis nana
B) Vibrio
C) Listeria
D) Norovirus

Answer: **A**

38. Aflatoxins and ochratoxin are:

A) Mycotoxins
B) Heavy metals
C) Radioactive nuclides
D) Viruses

Answer: **A**

39. Lead, cadmium and mercury cause:

A) Neurological and kidney damage
B) Only fever
C) Only rash
D) Only hearing loss

Answer: **A**

Noise and Lighting

40. Noise pollution is:

A) Pleasant sound
B) Unwanted or excessive sound harmful to health and environment
C) Silence
D) Only music

Answer: **B**

41. Permissible noise level for 8 hours:

A) 80 dB
B) 85 dB
C) 90 dB
D) 100 dB

Answer: **B**

42. Permissible exposure at 90 dB:

A) 16 h
B) 8 h
C) 4 h
D) 1 h

Answer: **C**

43. Permissible exposure at 110 dB:

A) 0.25 h
B) 1 h
C) 2 h
D) 8 h

Answer: **A**

44. Permissible level for 16 hours:

A) 80 dB
B) 85 dB
C) 95 dB
D) 110 dB

Answer: **A**

45. Which is NOT a harmful effect of noise?

A) Hearing loss
B) Hypertension
C) Disturbed sleep
D) Improved relaxation

Answer: **D**

46. Which is NOT an effect of weak lighting?

A) Headache
B) Eye pain
C) Photophobia
D) Cataract

Answer: **D**

47. Which is an effect of strong lighting?

A) Cataract
B) Muscle relaxation round the eye
C) Nystagmus
D) Pupil dilation

Answer: **A**

Radiation

48. Radiation does NOT include:

A) Alpha
B) Beta
C) Gamma rays
D) Sound waves

Answer: **D**

49. Acute radiation syndrome occurs at a whole-body dose typically:

A) Below 0.01 Sv
B) Above 1 Sv
C) 0 Sv
D) Below 0.1 Sv

Answer: **B**

50. Order of acute radiation stages:

A) Latent, prodromal, manifest
B) Prodromal, latent, manifest illness
C) Manifest, latent, prodromal
D) Latent, manifest, prodromal

Answer: **B**

51. Prodromal symptoms:

A) Nausea, vomiting, fatigue
B) Seizures and coma
C) Bone marrow failure
D) Cataract

Answer: **A**

52. Hematopoietic syndrome dose:

A) 1–6 Sv
B) 50 Sv
C) 0.01 Sv
D) 100 Sv

Answer: **A**

53. Hematopoietic syndrome mainly damages:

A) Bone marrow
B) Intestine
C) Brain
D) Skin only

Answer: **A**

54. Gastrointestinal syndrome dose:

A) 1 Sv
B) 6 Sv
C) 50 Sv
D) 0.1 Sv

Answer: **B**

55. Neurovascular syndrome dose:

A) 6 Sv
B) 10 Sv
C) 50 Sv
D) 1 Sv

Answer: C

56. Neurovascular syndrome leads to death within:

A) 48 hours
B) 1 year
C) 1 month
D) 10 years

Answer: A

Vectors & Parasites (House Fly, Ticks, Sand Fly, Lice, Mosquitoes)

57. Scientific name of the house fly:

A) Musca domestica
B) Culex pipiens
C) Aedes aegypti
D) Pediculus humanus

Answer: A

58. House fly order:

A) Diptera
B) Hemiptera
C) Anoplura
D) Acari

Answer: A

59. House fly family:

A) Culicidae
B) Muscidae
C) Ixodidae
D) Argasidae

Answer: B

60. House fly class:

A) Arachnida
B) Insecta
C) Mammalia
D) Reptilia

Answer: **B**

61. The house fly mainly acts as:

A) Mechanical vector
B) Biological reservoir of bacteria
C) Secondary host
D) None

Answer: **A**

62. Hard and soft ticks belong to superfamily:

A) Ixodoidea
B) Muscoidea
C) Culicoidea
D) Phlebotominae

Answer: **A**

63. Hard ticks family:

A) Ixodidae
B) Argasidae
C) Muscidae
D) Culicidae

Answer: **A**

64. Soft ticks family:

A) Ixodidae
B) Argasidae
C) Muscidae
D) Pediculidae

Answer: **B**

65. In soft ticks, the mouthparts:

A) Cannot be seen from above
B) Project forward prominently
C) Are absent
D) Are covered by scutum

Answer: **A**

66. Bulls-eye rash is called:

A) Erythema migrans
B) Photophobia
C) Nystagmus
D) Nits

Answer: **A**

67. Ticks spread:

A) Lyme disease and Rocky Mountain spotted fever
B) Cholera
C) Malaria
D) Polio

Answer: **A**

68. Sand flies belong to subfamily:

A) Phlebotominae
B) Culicinae
C) Anophelinae
D) Ixodinae

Answer: **A**

69. Sand flies are primary vectors of:

A) Leishmaniasis
B) Dengue
C) Cholera
D) Typhoid

Answer: **A**

70. Carrion's disease is caused by:

A) Bartonella bacilliformis
B) Vibrio cholerae
C) Listeria
D) Giardia

Answer: **A**

71. Carrion's disease is transmitted by genus:

A) Lutzomyia
B) Aedes
C) Culex
D) Musca

Answer: **A**

72. Body lice live in:

A) Seams and folds of clothing
B) Water
C) Soil
D) Food

Answer: **A**

73. Lice eggs are called:

A) Nits
B) Larvae
C) Tumblers
D) Wrigglers

Answer: **A**

74. Body lice are treated with lotion/shampoo containing:

A) 1% permethrin
B) 50% alcohol
C) Chlorine
D) Aluminum sulphate

Answer: **A**

75. Anopheles transmits:

A) Malaria
B) Cholera
C) Typhoid
D) Lyme disease

Answer: **A**

76. Culex causes filariasis and carries:

A) Viral encephalitis
B) Yellow fever only
C) Malaria
D) Dengue

Answer: **A**

77. Aedes carries pathogens of:

A) Yellow fever, dengue, Zika
B) Malaria only
C) Cholera
D) Lyme disease

Answer: **A**

78. Mosquito larvae are called wrigglers, and pupae are called:

A) Nits
B) Tumblers
C) Nymphs
D) Cysts

Answer: **B**

Part 2: True / False

1. Ventilation only means replacing stale air.

Answer: **False** (it also controls quality of incoming air)

2. Solid waste can cause skin and gastro-intestinal diseases.

Answer: **True**

3. Burning waste creates toxic ash.

Answer: **True**

4. Water has two hydrogen atoms and one oxygen atom.

Answer: **True**

5. Drinking water turbidity should not exceed 5 NTU.

Answer: **True**

6. A pH above 9 improves chlorination.

Answer: **False** (it constrains it)

7. Zero coliform means the water is fit.

Answer: **True**

8. 10–100 E. coli means contaminated.

Answer: **True**

9. Minimum water requirement is 15 L/person/day.

Answer: **True**

10. One tap may serve up to 250 people.

Answer: **True**

11. One hand pump may serve up to 250 people.

Answer: **False** (500)

12. 50 people per bathing facility is the maximum.

Answer: **True**

13. Aluminum sulphate is used in water treatment.

Answer: **True**

14. Cholera and typhoid are related to drinking-water contamination.

Answer: **True**

15. Foodborne illnesses can be caused by chemicals.

Answer: **True**

16. Listeria can grow at refrigeration temperatures.

Answer: **True**

17. Norovirus is a bacterium.

Answer: **False** (virus)

18. Hepatitis A can spread through contaminated raw produce.

Answer: **True**

19. Aflatoxins are produced on grain.

Answer: **True**

20. Noise pollution is unwanted or excessive sound.

Answer: **True**

21. Permissible noise for 4 hours is 90 dB.

Answer: **True**

22. Permissible exposure time decreases as noise level increases.

Answer: **True**

23. Hypertension is a harmful effect of noise.

Answer: **True**

24. Strong lighting may cause cataract.

Answer: **True**

25. Gamma rays are electromagnetic waves.

Answer: **True**

26. Acute radiation syndrome occurs at 0.01 Sv.

Answer: **False** (above 1 Sv)

27. The latent period is when symptoms subside but cellular damage continues.

Answer: **True**

28. Hematopoietic syndrome damages bone marrow.

Answer: **True**

29. Neurovascular syndrome has an effective treatment.

Answer: **False** (none)

30. Musca domestica belongs to order Diptera.

Answer: **True**

31. House flies are a secondary host of medically important bacteria.

Answer: **False** (mechanical carriers)

32. Ixodidae are hard ticks.

Answer: **True**

33. Argasidae are hard ticks.

Answer: **False** (soft ticks)

34. Erythema migrans is a bulls-eye rash in Lyme disease.

Answer: **True**

35. Both male and female sand flies suck blood.

Answer: **False** (only the female)

36. Sand flies are vectors of leishmaniasis.

Answer: **True**

37. Body lice lay eggs in the seams of clothing.

Answer: **True**

38. Permethrin products are used to treat lice.

Answer: **True**

39. Culicidae is the mosquito family.

Answer: **True**

40. Only female mosquitoes need blood meals to mature their eggs.

Answer: **True**

41. Aedes transmits yellow fever, dengue and Zika.

Answer: **True**

42. Mosquito eggs are laid on the surface of water.

Answer: **True**